3.2.3 Controlling communicable diseases

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the principles of vaccination	To include the different forms of vaccines (live vaccine, dead microorganisms, pathogen fragments) and the importance of booster vaccinations.
(b)	the role of vaccination programmes in the prevention of epidemics	To include reference to the establishment of herd immunity.
(c)	the biological problems in the development of vaccines and the use of vaccination programmes	To include issues with vaccine development, mutation rate and antigen variability (e.g. in HIV and the influenza virus(es)) and live vaccines **AND** vaccine use – storage of vaccine, distribution of vaccine and the nutritional status of the target population e.g. if protein deficient.
(d)	the ethical issues related to the development and use of vaccines	To include the use of a vaccine in girls against Human Papilloma Virus (HPV) to prevent cervical cancer. HSW10
(e)	the use of antibiotics in the treatment of communicable disease	To include an outline of the modes of action of antibiotics e.g. inhibition of bacterial protein, DNA and cell wall synthesis **AND** the cellular differences between prokaryotic and eukaryotic cells that allow antibiotics to act on bacterial but not human cells.
(f)	how the misuse of antibiotics can lead to the evolution of resistant strains of bacteria	To include reference to TB and MRSA.
(g)	practical investigation on the effect of antibiotics on Gram-positive and Gram-negative bacteria.	To include bacteriostatic and bacteriocidal effects of antibiotics and the effects of disinfectant use and other hygiene practices. *M0.1, M0.2, M0.4, M1.1, M4.1* **PAG1, PAG7** HSW3, HSW4, HSW5, HSW6, HSW8